BlinkIT — Grocery Sales & Outlet Performance Analytics

Data Analyst Capstone Project | Uday Dubey | Tools: **Excel** + **Power BI** | Source: BlinkIT-Grocery-Data, 8,523 item-outlet records

EXCEL POWER BI

1. Business Problem

BlinkIT operates 10 outlets across Tier 1-3 cities, stocking 1,559 items across 16 categories. Category Management had no consolidated view of what actually drives sales — decisions on assortment and outlet expansion were being made on gut feel and manager anecdotes. The VP of Merchandising needed evidence-backed answers on outlet formats, category performance, item ratings, and outlet age.

2. Data & Cleaning (Excel)

Issue found	Action taken
Fat content had 5 inconsistent labels for 2 real categories	Standardized to exactly 2: Low Fat / Regular
Item weight missing on 1,463 rows (17%)	Backfilled from same item elsewhere, then category median
Item visibility = 0 on 526 rows (physically impossible)	Treated as missing data, not a literal zero
Outlets established as recently as 2022	Outlet age computed from a fixed 2023 reference year

$1,20,168	10	1,559	3.92 / 5
Total Sales	Outlets	Items	Avg Rating

3. Analysis & Dashboard (Power BI)

Built a 3-page interactive Power BI dashboard — **Executive Summary** (KPIs, sales by item/outlet type, location split, year trend), **Product & Outlet Analysis** (visibility-vs-sales scatter, fat content, outlet size), and **Outlet Performance** (per-outlet sales & rating, trend by year, full outlet x item-type table), all driven by DAX measures. Used it to answer four questions:

1. **Which outlet formats & locations drive sales?**
   Supermarket Type1 generates **65.5%** of total sales ($7,87,550 of $12,01,682) from just 6 of 10 outlets. **Tier 3** cities lead ($4,72,133) over Tier 2 and Tier 1. Six outlets cluster near $1.3L each; four — notably OUT010, OUT019 — lag at $47K-$78K.
2. **Which categories over/under-perform their shelf space?**
   A sales-share vs visibility-share index shows **Household** (1.15x) and **Health & Hygiene** (1.11x) convert shelf space best; **Breakfast** (0.78x) and **Baking Goods** (0.85x) underperform their space.
3. **Does item rating predict sales?**
   No — correlation is **0.011** (rating) and **0.001** (visibility), both negligible. Category mix and outlet format drive sales far more than either.
4. **How does outlet age affect performance?**
   Weak negative correlation (**-0.25**) — newer outlets hold their own against older ones, a good sign for continued expansion.

4. Recommendations

1. Prioritize **Supermarket Type1 in Tier 3 cities** for the next round of expansion.
2. Reallocate shelf space from **Breakfast** and **Baking Goods** toward **Household** and **Health & Hygiene**.
3. Stop treating rating and visibility as merchandising KPIs — focus review effort on category mix and outlet-format fit instead.

Deliverables: cleaned .xlsx workbook (Excel) · .pbix dashboard (Power BI) · this insight summary. All figures computed directly from the cleaned dataset.

blinkit
SALES ANALYSIS REPORT

Total_Sales	Avg_Rating	Total_Outlets	Total_items
1.20M	3.92	10	34

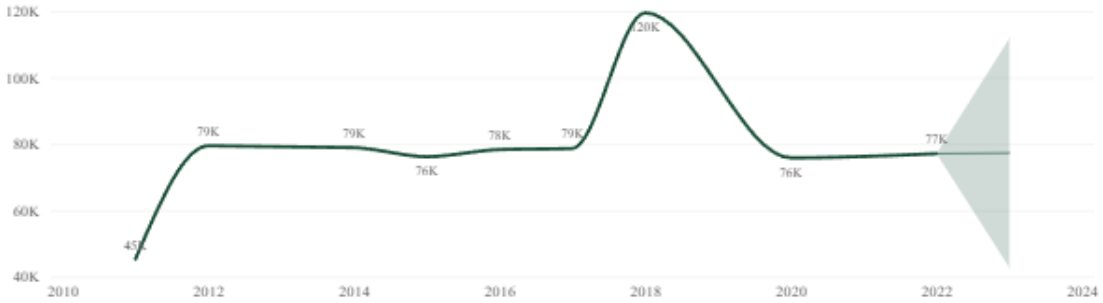
Outlet_Size
- [ ] High
- [ ] Medium
- [ ] Small

Outlet_Establishment_Year
- [ ] 2015
- [ ] 2016
- [ ] 2017
- [ ] 2018
- [ ] 2020
- [ ] 2022

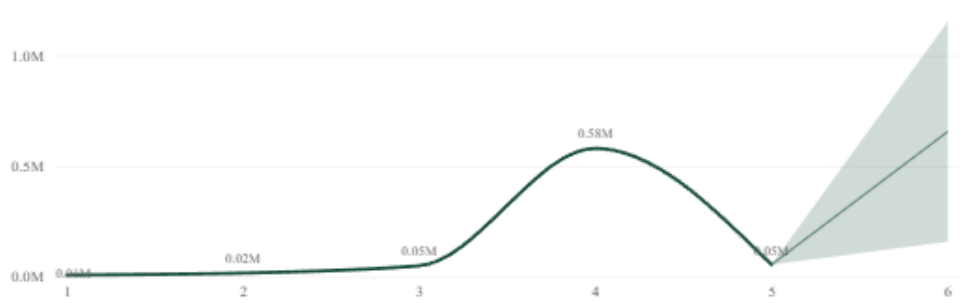
Outlet_Location_Type
- [ ] Tier 1
- [ ] Tier 2
- [ ] Tier 3

Item_Fat_Content
- [ ] Low Fat
- [ ] Regular

Total Sales by Outlet Establishment Year
Total_sales by Outlet_Establishment_Year



Total sales by Rating
Total_sales by Rating



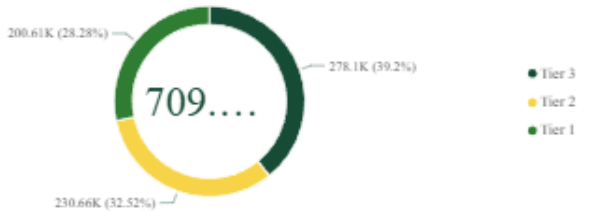
Top Performing Product Categories
Total_sales by Item_Type



Total Sales by Outlet Type
Total_sales by Outlet_Type



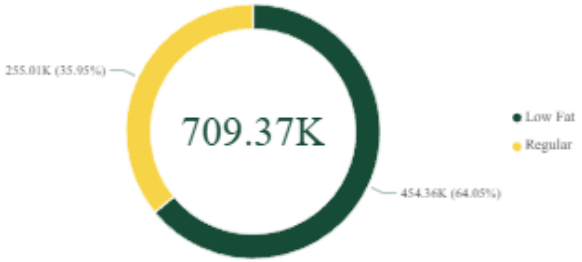
Total Sales by Outlet Location Type
Total_sales by Outlet_Location_Type



Avg_Rating	Average_Visibility	Average_Weight	Total_categories
3.92	0.07	10.65	16

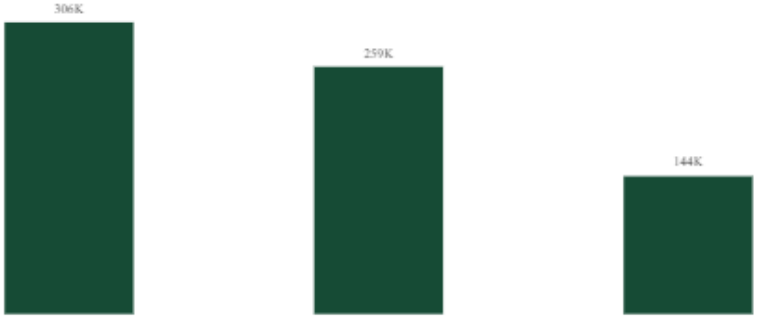
Total Sales by Item Fat Content

Total_sales by Item_Fat_Content



Total Sales by Outlet Size

Total_sales by Outlet_Size



Outlet_Location_Type

- Tier 1
- Tier 2
- Tier 3

Outlet_Type

- Grocery Store
- Supermarket Type1
- Supermarket Type2
- Supermarket Type3

Outlet_Size

- High
- Medium
- Small

Outlet_Establishment_Year

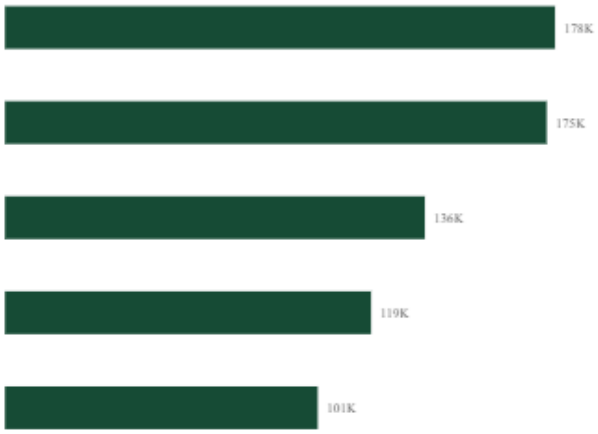
- 2011
- 2012
- 2014
- 2015
- 2016
- 2017
- 2018

Item_Fat_Content

- Low Fat
- Regular

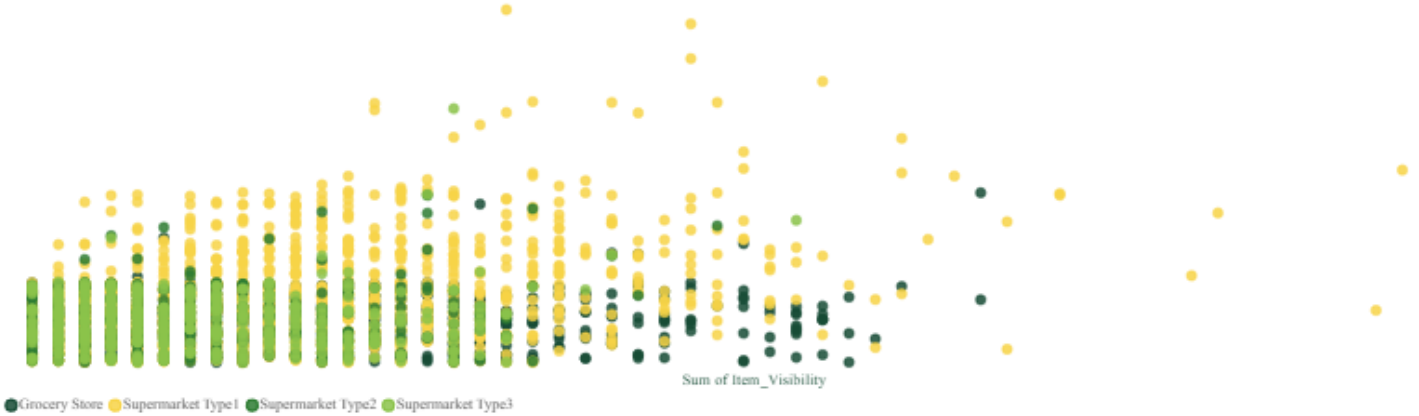
Total Sales by Item Type

Total_sales by Item_Type



Sales vs. Item Visibility by Outlet Type

Sum of Item_Visibility, Sum of Sales and First Item_Type by Sales and Outlet_Type



Avg Sales Per Outlet	Avg_Rating	Total_Outlets	Total_Categories
120.17K	3.92	10	16

Outlet_Location_Type
- Tier 1
- Tier 2
- Tier 3

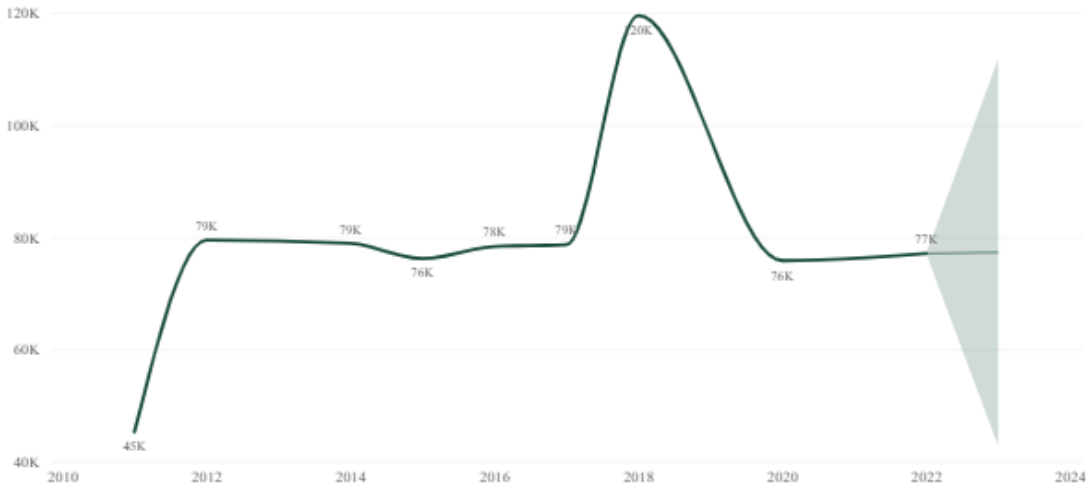
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Outlet_Type
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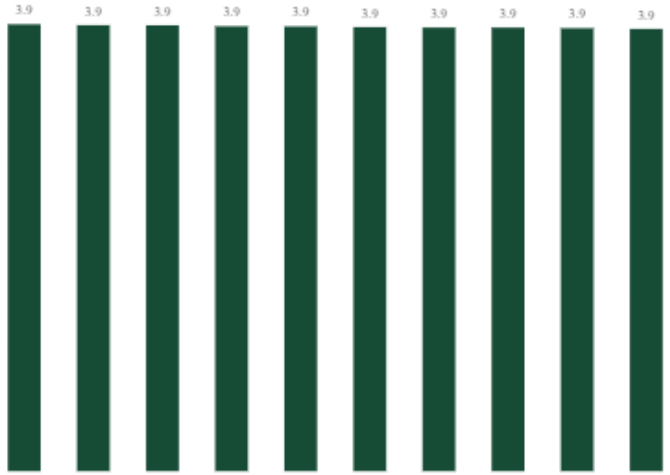
Total Sales by Outlet Establishment Year

Total_sales by Outlet_Establishment_Year



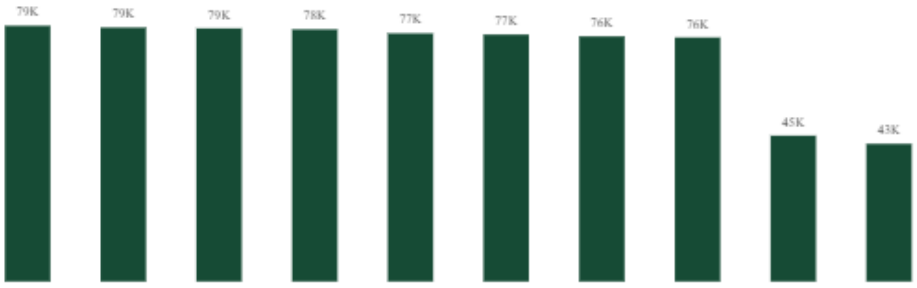
Average Rating by Outlet Identifier

Avg_rating by Outlet_identifier



Total Sales by Outlet Identifier

Total_sales by Outlet_identifier



Outlet & Item Category Performance Analysis

Outlet_identifier	Item_Type	Total_sales	Avg_sales_per_outlet	Total_items	Avg_rating	Avg_visibility	Avg_Weight	Outlet_Size
OUT010	Snack Foods	674.60	674.60	6	4.33	0.09	14.06	Medium
OUT017	Frozen Foods	558.14	558.14	3	4.33	0.09	12.52	Medium
OUT017	Dairy	1,761.56	1,761.56	8	4.09	0.06	11.69	Medium
OUT010	Dairy	4,298.01	4,298.01	16	4.08	0.11	14.57	Medium
OUT010	Snack Foods	1,991.02	1,991.02	10	4.08	0.08	15.62	Small
OUT019	Household	8,260.03	8,260.03	19	4.06	0.11	0.00	Small
OUT046	Household	15,436.05	15,436.05	19	4.04	0.06	13.73	Small
OUT045	Household	12,509.80	12,509.80	18	4.04	0.05	13.32	Small
OUT010	Frozen Foods	4,508.72	4,508.72	15	4.03	0.07	12.62	Medium
Total		**709,370.85**	**70,937.09**	**34**	**3.92**	**0.07**	**10.92**	